{EPITECH}

TRINITY - PART THREE

< APP DEV />

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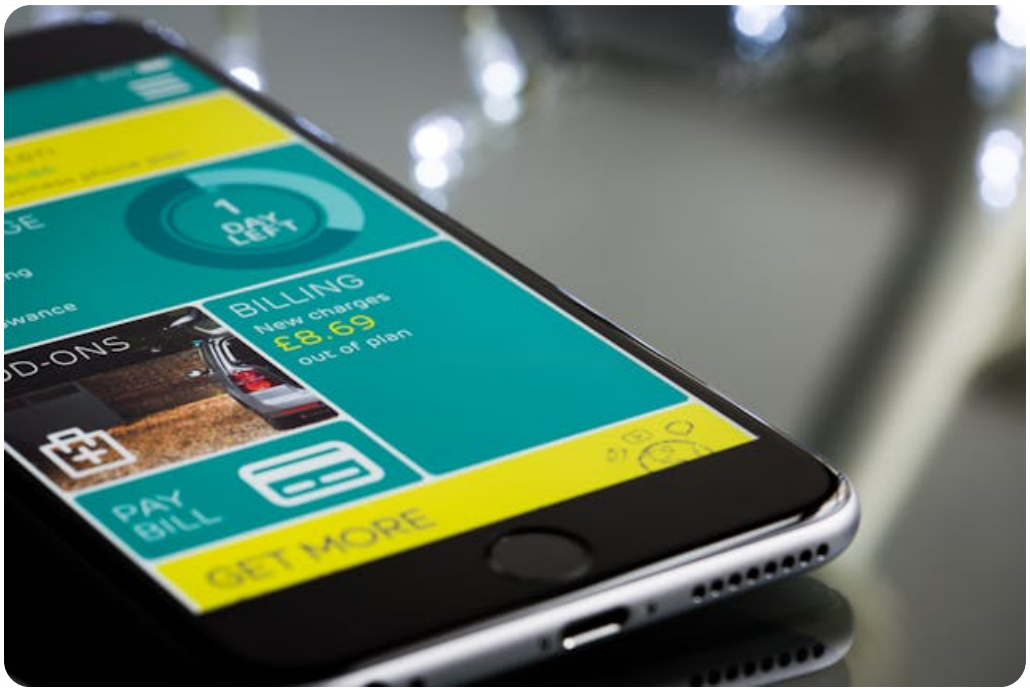
This project is composed of **three distinct parts**: dev Ops, dev Web and dev App.
The **timing** of deliveries and presentations has already been **fixed**.

To boost competitiveness, a grocery chain needs your dev skills to achieve several goals:

- ✓ They want to improve the customers' user experience, simplify their purchasing process, and enhance their satisfaction. To achieve this, you are tasked with developing an integrated and intuitive **mobile** application.
- ✓ Internally, they aim to optimize product management, streamline the supervision of sales, and facilitate decision-making through the visualization of key performance indicators. You are therefore assigned to implement a **web** application connected to a RESTful API.

Application Development

You're finally tasked to develop an intuitive mobile application that streamlines the purchasing process for end customers, improve their experience and enhance their satisfaction.



This application not only allows users to scan products and manage shopping carts, but also to access detailed product information and complete payments directly from their mobile devices.

The main features of this mobile app will include, at least:

- ✓ a login screen;
- ✓ a home page;
- ✓ a scanner;
- ✓ product display;
- ✓ cart management;
- ✓ a payment procedure;
- ✓ purchase history.

You can use any tool or language you fancy (java, kotlin, swift, swiftUI, react-native, flutter, ...).
Be aware that you'll be asked to expose and justify your choices.

Specification details

Main features

✓ **Login screen:**

- allow users to log in via email and password;
- provide the option to sign up and create an account;
- implement a secure login process.

✓ **Home page:**

- present an overview of the application with an attractive and intuitive design;
- display clear navigation buttons to quickly access main features:
  - scan a product,
  - view the cart,
  - access purchase history,
  - manage the account;
- include promotional sections or product recommendations based on the user's purchase history;
- show notifications or important messages (e.g., current promotions).

✓ **Scanner:**

- use the smartphone's camera to scan the products' bar codes;
- handle scan errors (e.g., unreadable bar codes, products not found) and provide appropriate feedback to users.

✓ **Product display:**

- allow users to add products to the cart after scanning them;
- display detailed information of scanned products, including at least:
  - name,
  - picture,
  - brand,
  - category,
  - nutritional information,
  - price,
  - available quantity in stock.

This list is **non-exhaustive**: you can add as many fields as you like, keeping in mind the size of your screen and ensuring a clear display for the user.

✓ **Cart management:**

- display the contents of the cart with the ability to modify quantities or remove items;
- calculate and display the total purchases in real-time.

✓ **Proceed to payment:**

- allow users to enter billing information via a form:
  * first name,
  * last name,
  * address,
  * zip code,
  * city;
- **mandatory integration of the PayPal API** to enable secure payment for purchases

✓ **Purchase history:**

- display the user's previous purchase history with transaction details.

Technical challenges

✓ **Camera usage:**

- camera access permissions are secure and respectful, adhering to user privacy;
- ensure a smooth and latency-free UX by optimizing the app performance;
- process images in real-time to quickly decode bar codes, even in low-light conditions.

✓ **Transaction security:**

- ensure secure communication between the app and the payment services, via HTTPS and robust encryption protocols;
- implement verification and fraud protection mechanisms for financial transactions (e.g., transaction callback for payment verification).

✓ **API integration:**

- use the internal API to manage user data, products, and transactions;
- implement secure requests using JWT to authenticate and authorize users;
- use caching mechanisms to reduce redundant API calls and improve performance.

✓ **User experience and user Interface:**

- design an intuitive and friendly interface to ease the use of your application;
- follow UX/UI best practices to create a smooth and frictionless experience;
- conduct user testing to identify and address usability issues;
- use attractive visuals and animations to enhance interaction with your application.

✓ **Code architecture:**

- adopt a modular architecture to allow for better maintainability and scalability;
- use design patterns like MVVM or MVP to separate concerns and facilitate testing;
- implement reusable services and components to optimize development;
- document the code architecture and data flows to ease onboarding for new developers and enhance team collaboration.

✓ **Testing:**

- write some unit tests for all critical application features, to ensure their proper functionality and guarantee at least 20% of code coverage;
- automate the unit tests as part of the CI pipeline to quickly detect regressions;
- use testing frameworks (JUnit for Java/Kotlin or XCTest for Swift).

Delivery and keynote

During the final keynote, you are expected to:

- ✓ present the whole project and sum up the 3 parts;
- ✓ present and justify your mobile application;
- ✓ demonstrate, on a smartphone (android or iOS), its proper functioning, including the links with the APIs (yours and the one from PayPal);
- ✓ identify the challenges encountered and overcome;
- ✓ present the potential areas for further improvement.

> ⚠ A demonstration via an emulator or web will NOT be accepted.

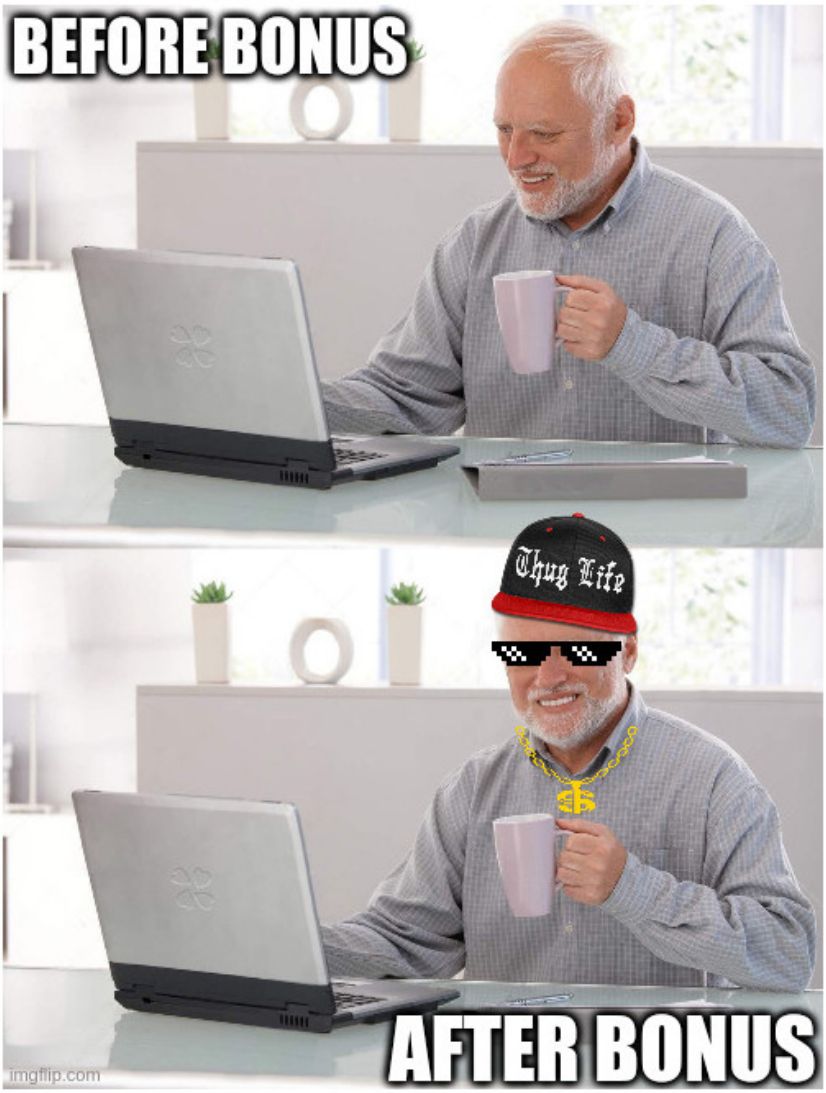
For this specific part, you are expected to deliver:

- ✓ The source code of your mobile application.
- ✓ The technical documentation for:
  - its architecture and its components;
  - your technological choices;
  - the data flows describing how data moves within the application.
- ✓ Some UML diagrams, including:
  - a class diagram showing data structures and their relationships;
  - an activity diagram, illustrating different workflows and business processes.
- ✓ The unit tests report, which should be directly visible in the CI/CD pipeline.

Bonuses

Once you finished the mandatory parts, add some optional features to your project, such as:

- ✓ provide an option to send purchase receipts via email directly from the application;
- ✓ add other payment options, such as the local currency for your region;
- ✓ implement a guided tour of your mobile application;
- ✓ make the report customizable, with awesome dataviz for your KPIs;
- ✓ include modes to improve the accessibility of your interfaces.



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